

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com

Arduino IDE, select File→Preferences and paste the link given below.

http://arduino.esp8266.com/stable/package_esp8266com_index.json

Open Board Manager from Tools→Board Menu and install ESP8266 platform. (Note: Do not forget to select your ESP8266 board from Tools→Board menu after installation.)

In this project, external header files are needed for programming. This is a simple way to detect the joystick, and compare x and y values from Blynk app using simple programming functions.

Construction and testing

Wire the circuit on the breadboard along with NodeMCU. Connect the left and right motors on the chassis of the robot. Fix a suitable castor wheel at front of the chassis.

In this project four digital I/O pins

(D1 and D2, D5 and D6) of NodeMCU are connected to inputs of L293D H-bridge motor driver IC for controlling right and left motors of the robot.

Case 1. When you move the joystick with your finger in x-negative direction on the joystick panel, digital value is less than 30. In this condition, voltage at digital pin D6 is high, and voltages at digital pins D5, D1 and D2 are low. In this condition, the robot moves in left direction till you release the button on joystick panel.

Case 2. When you move the joystick in x-positive direction, digital value is less than 220, and voltages at pins D6, D5 and D1 are low, while at pin D2 it is high. In this condition, the robot moves in right direction till you release the button on joystick panel.

Case 3. When you move the joystick in y-negative direction, digital value is less than 30, voltages at pins D6 and D2 are high, while at pins D5 and D1 are low. In this condition, the robot moves in reverse/backward direction till you release the button on joystick panel.

Case 4. When you move the joystick in y-positive direction, digital

PARTS LIST	
<i>Semiconductors:</i>	
Board1	- NodeMCU ESP8266 board
IC1	- 7805, 5V voltage regulator
IC2	- L293D motor driver
<i>Capacitors:</i>	
C1	- 100μF, 35V electrolytic
C2	- 0.1μF ceramic disk
<i>Miscellaneous:</i>	
BATT.1	- 12V battery
M1-M2	- 12V DC geared motor
	- Chassis
	- Wheels for rear motors
	- Castor wheel for front

value is less than 220, voltages at pins D6 and D2 are low, while at pins D5 and D1 are high. In this condition, the robot moves in forward direction till you release the button on joystick panel.

Case 5. Joystick is in neutral condition. Digital values for x and y directions are 128. Here, voltages at all digital pins (D6, D5, D2 and D1) are low. In this condition, the robot is in stop condition. **EFY**



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